

Graphs of Functions

- 1) Multiple choice: What is the name of the function whose graph looks like a series of steps?
 - A. odd function
 - B. even function
 - C. step function
 - D. absolute function
- 2) Identify the domain, range, and intervals where the function is increasing, decreasing, or constant.
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- 4) Given the graph of the function shown, identify the relative extrema.
- 5) The floor function is $h(x) = \lfloor x \rfloor$.
 - a) Evaluate $h(11.1)$.
 - b) Evaluate $h(9.7)$.
- 6) Graph $f(x) = -|x - 2| + 1$ using technology, then write the intervals where the function is increasing and decreasing.
- 7) Graph $g(x) = \frac{x^2}{3}$ using technology, then write the intervals where the function is increasing and decreasing.
- 8) Determine whether $f(x) = |x| + x^4 + 2x^2 + 1$ is odd, even, or neither. Show your work.
- 9) Determine whether $m(x) = \sqrt{x - 1}$ is odd, even, or neither. Show your work.
- 10) Laura and Marcus take their three children: Ava, Ethan, and Zoe, to the water park. The cost depends on age and is shown in the step function. Laura and Marcus are both 43 years old. Ava is 12, Ethan is 8, and Zoe is 6. What is the total cost for the family of five?
- 11) Use technology to find the relative maximum and minimum of $f(x) = x^3 - 3x^2 + 5$.
- 12) Show that the function $f(x) = x^3 - x$ is classified as odd.
- 13) A roller coaster's vertical position (in meters) over time (in seconds) is modeled by $h(x) = -0.2x^4 + 4x^3 - 24x^2 + 48x + 1$. Use graphing technology to determine when the roller coaster reaches high and low points.
- 14) Graph $f(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$.
- 15) Graph the step function based on the information below: -2 for $-7 \leq x < -4$, 1 for $-4 < x < -1$, 8 for $-1 \leq x < 1$, 5 for $1 < x < 4$, and 6 for $4 \leq x \leq 7$.
- 16) Write an equation for the step function shown.
- 17) Given the graphs below, determine whether the functions are odd, even, or neither.
- 18) Two students are debating whether $f(x) = \frac{x}{x^2 + 1}$ is odd. One says no because "the exponents are not all odd." The other says yes. Evaluate their reasoning and determine who is correct.

- 19) Write the equation of and graph a function that is increasing on $(-\infty, -2]$, constant on $[-2, 1]$, and decreasing on $[1, \infty)$.
- 20) Sketch a graph of a function that has one relative maximum and two relative minimums.